

DIFFERENTIATION WITH EXPONENTIALS AND LOGARITHMS

Complete the problems below. Solutions and marking available upon request.

1. Let $f(x) = 6e^x + 4$. Show that $f'(3) \approx 120.5132$ (4dp).
2. Let $g(x) = \ln(x^2 - 1)$. Show that $g'(3) = 0.6$.

3. The temperate T of a cup of tea is modelled to be

$$T = 85e^{-t/10} + 15,$$

where t is the time, in minutes, after the tea was first made.

- (a) What is the initial temperate T of the cup of tea?
- (b) How quickly is the tea cooling down after $t = 1$ minutes? How quickly is it cooling down after $t = 10$ minutes?
- (c) What is the coldest temperate this cup of tea will ever be?

Hint. Asking “how quickly is something cools down” can be interpreted as “what is the rate of change of the temperature” (of this object)?

4. The population P of seagulls in Brighton is modelled according to

$$P = \ln(5t + 10),$$

where $t = 0$ represents the year 1800, and P is measured in thousands.

- (a) According to this model, what was Brighton’s seagull population in the year 1800?
- (b) Find the derivative $\frac{dP}{dt}$. What does this quantity represent?
- (c) Evaluate $\frac{dP}{dt}$ for $t \in \{0, 1, 20, 100\}$.
- (d) Provide an interpretation for the different values in (c).

Hint. The notion of a ‘carrying capacity’ from biology is relevant here.

5. Consider the two functions

$$\begin{aligned} f(x) &= e^{\sin(x-2)+1/5x-2/5}, \\ g(x) &= \ln(x^3 + 2). \end{aligned}$$

Show that $f'(x) = g'(x)$ at $x = 2$.